

The Goalkeeper in Possession

Visible Options, Retention Risk, and Asymmetric Turnover Cost

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Introduction. Goalkeeper distribution is usually evaluated after the pass is played, yet coaches decide before it: build short, play through pressure, or go direct. We ask whether the options visible to a goalkeeper explain those choices and improve retention modeling, and how that probability should be interpreted when failure near goal is disproportionately costly.

Methods. We analyze 45,974 goalkeeper distributions from 722 StatsBomb open-data matches across five elite cohorts. A target-free expected-retention model (xR-GK) is evaluated with true leave-one-competition-out testing and independent out-of-fold value models. We additionally analyze 4,831 goalkeeper distributions with StatsBomb 360 freeze frames from the 2022 World Cup and Euro 2024. Visible teammates are classified by distance, centrality, receiver pressure, and whether a 1.5-m passing corridor is clear. On 4,802 exact joins to frozen xR-GK, five-fold match-grouped models test whether two 360 variables - clear short-option count and a visible central short exit - add information beyond locally recalibrated xR, event pressure, goal-kick state, and competition.

Results. Direct play was observed on 85.9% and 91.9% of distributions with no clear short exit in the World Cup and Euro samples, respectively, falling to 27.2% and 25.8% with two or more clear short exits; the pattern survived completed-pass and lane-width sensitivities. Adding the two 360 variables improved match-held-out AUC from 0.648 to 0.699 (delta=0.051, 95% CI 0.035-0.066), Brier score by 0.0065, and log loss by 0.0167, with the same direction in both tournaments and after visible-area control. Event-only xR retained moderate discrimination in unseen competitions (AUC 0.645-0.710) but absolute probabilities shifted materially. Independently refit cohort models also showed that opponent consequence after failure exceeded the origin possession value in every environment (C/V 1.40-1.90), making the complete lost branch 2.40-2.90 times origin value.

Conclusion. Goalkeeper distribution is an option-constrained decision under asymmetric risk. The visible option set explains both short-versus-direct behavior and retention beyond event-only controls, while probability scale remains environment-dependent and failure near goal is systematically costly. For clubs, the practical task is not to assign one universal “good with his feet” score, but to ask what options the goalkeeper sees, how likely each distribution is to retain possession in the relevant environment, and what the team risks if it fails.

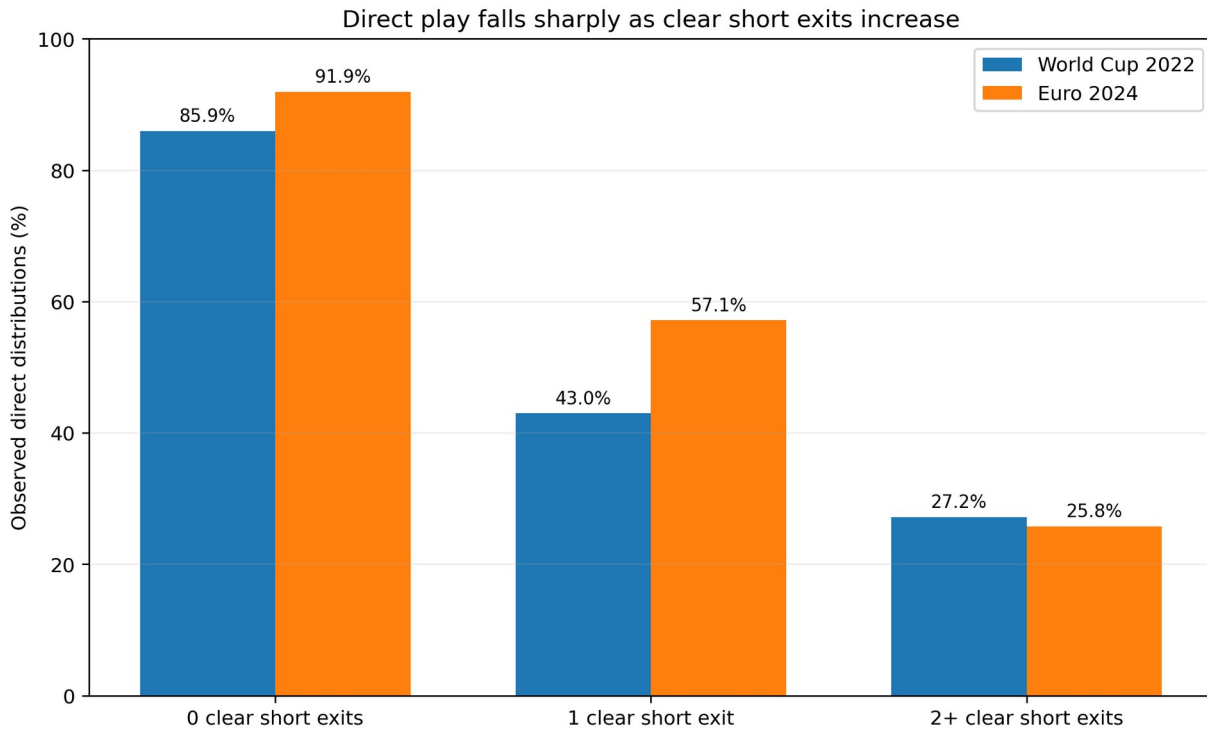


Figure A. Observed direct distributions decrease sharply as the number of visible lane-clear short exits rises in both World Cup 2022 and Euro 2024.

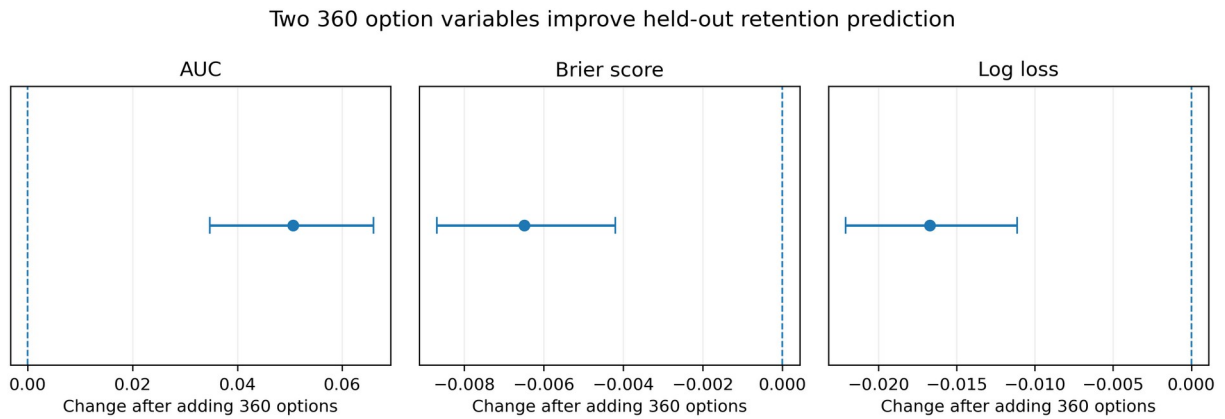


Figure B. On 4,802 exact event joins, two fixed 360 option variables improve match-held-out retention prediction beyond locally recalibrated xR and non-360 controls. Error bars are 95% paired match-cluster bootstrap intervals.

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